

Message Vector

$$\mathbf{u}$$
$$1 \times k$$

$\times$

Generator Matrix $\mathbf{G}$

$$\left[\begin{array}{c|c} \mathbf{I}_k & \mathbf{A} \end{array} \right]$$

$k \times k$ $k \times (n - k)$

$=$

Systematic Codeword $\mathbf{c} = \mathbf{uG}$

$$\left[\begin{array}{c|c} \mathbf{u} & \mathbf{uA} \end{array} \right]$$

$1 \times k$ $1 \times (n - k)$

Information Bits

Parity Bits